

CO3 AT3-Comparative Analysis

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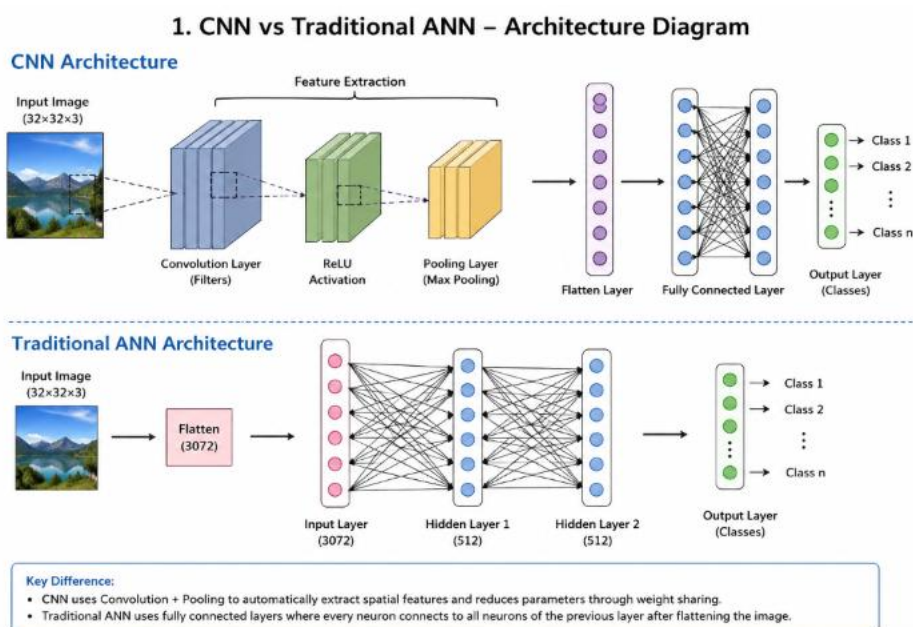
1. CNN vs Traditional Neural Networks

Explanation

A Traditional Artificial Neural Network (ANN) consists mainly of fully connected layers where every neuron is connected to neurons in the next layer. For image data, the image must usually be flattened into a one-dimensional vector, which results in a very large number of parameters. ANNs do not naturally preserve the spatial relationship between neighboring pixels.

A Convolutional Neural Network (CNN) is specially designed for image processing. It uses convolution filters to automatically extract features such as edges, textures, shapes, and objects. CNNs use local connectivity and parameter sharing, which significantly reduces the number of parameters and computational requirements compared with fully connected networks.

ARCHITECTURE DIAGRAM



COMPARISON TABLE

Comparison Aspect	CNN	Traditional ANN
Architecture	Convolution, pooling and fully connected layers	Mainly fully connected layers
Parameter Sharing	Uses shared convolution filters	No parameter sharing
Feature Extraction	Automatically extracts spatial features	Usually requires flattened input
Spatial Information	Preserves spatial relationships	Spatial relationships are mostly lost
Computational Complexity	Fewer parameters and efficient for images	Very high parameters for large images
Image Processing	Highly suitable for image processing	Less suitable for image processing

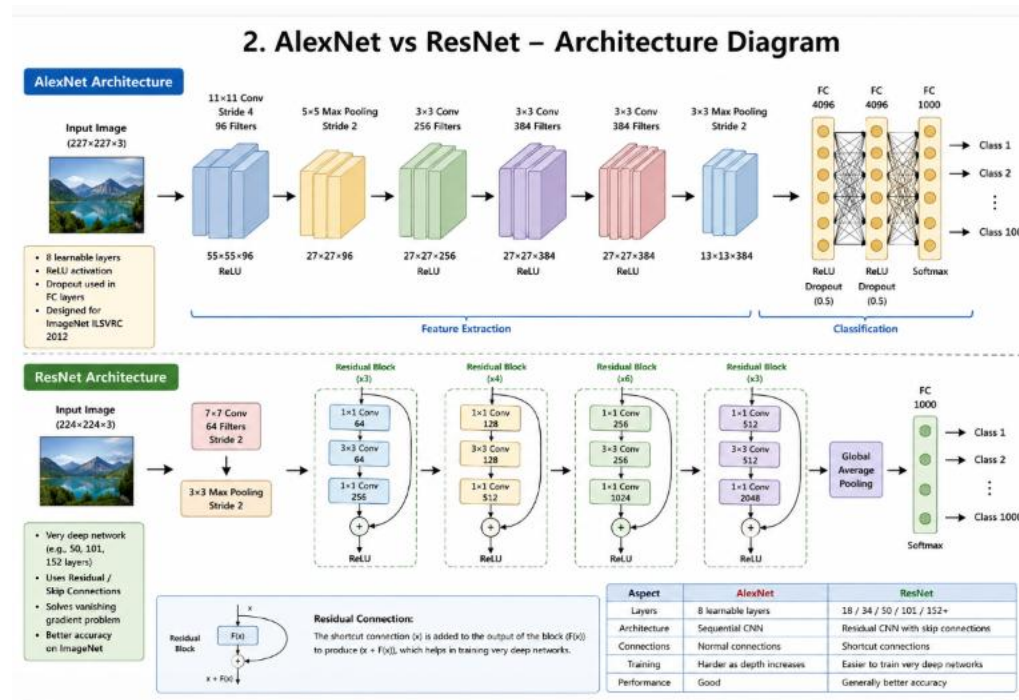
2. AlexNet vs ResNet

Explanation

AlexNet is a pioneering deep CNN architecture that became famous for its strong performance in the ImageNet competition. It contains 8 learnable layers and uses techniques such as ReLU activation, dropout, data augmentation, and GPU-based training. However, as networks become deeper, training becomes more difficult because of vanishing gradients and degradation problems.

ResNet (Residual Network) solves this problem using residual/skip connections, which allow information and gradients to pass directly through layers. ResNet can contain 18, 34, 50, 101, or 152 layers while remaining easier to optimize. As a result, ResNet generally achieves better accuracy and allows much deeper architectures than AlexNet.

ARCHITECTURE DIAGRAM



COMPARISON TABLE

Comparison Aspect	AlexNet	ResNet
Architecture	Conventional deep CNN	CNN with residual blocks
Number of Layers	8 learnable layers	18, 34, 50, 101, 152+ layers
Connections	Sequential connections	Skip/residual connections
Training	More difficult as network depth increases	Easier to train deep networks
Computational Requirement	High GPU requirement	Higher for deeper versions
Performance	Good image classification performance	Generally higher accuracy

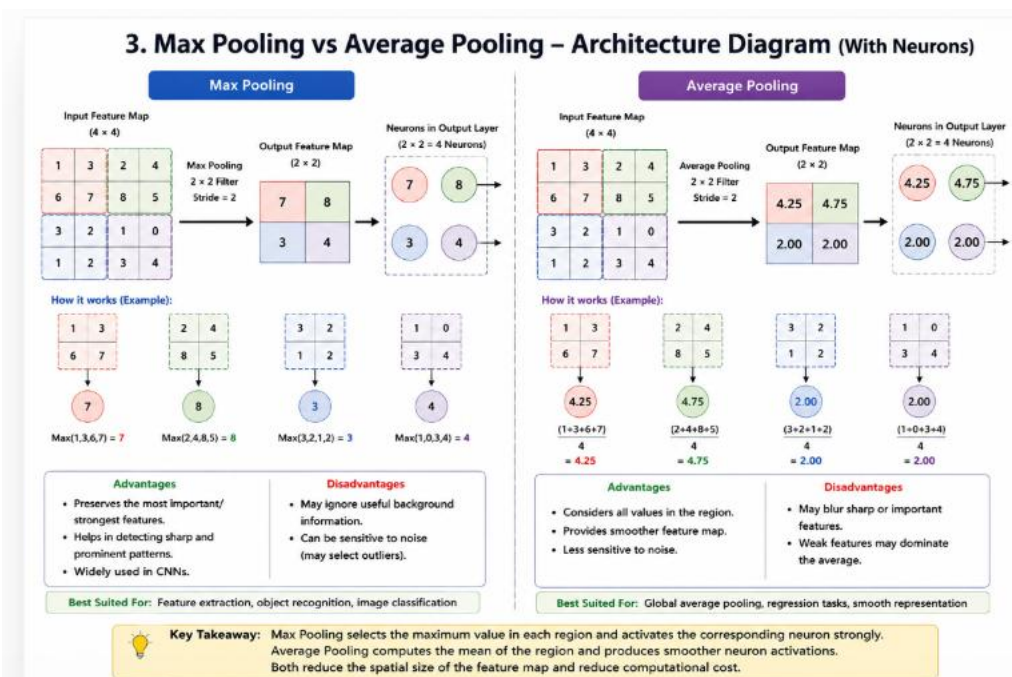
3. Max Pooling vs Average Pooling

Explanation

Max Pooling selects the maximum value from a particular region of a feature map. For example, if a 2×2 region contains $[1, 3; 2, 4]$, max pooling produces 4. It preserves the strongest and most important activation, making it useful for detecting prominent features such as edges and object parts.

Average Pooling calculates the average of all values in the selected region. For the same example, the output is $(1+3+2+4)/4 = 2.5$. Average pooling provides smoother feature representations but may reduce strong feature information. Max pooling is commonly used in CNN feature extraction, while average pooling is often used for reducing feature maps and in global average pooling.

ARCHITECTURE DIAGRAM



COMPARISON TABLE

Comparison Aspect	Max Pooling	Average Pooling
Working	Selects the maximum value	Calculates the average value
Example	$[1,3;2,4] \rightarrow 4$	$[1,3;2,4] \rightarrow 2.5$
Feature Preservation	Preserves strongest features	Preserves overall information
Output	Sharper feature representation	Smoother representation
Advantage	Good at detecting prominent features	Provides smooth feature representation
Application	Common in CNN feature extraction	Often used for global average pooling

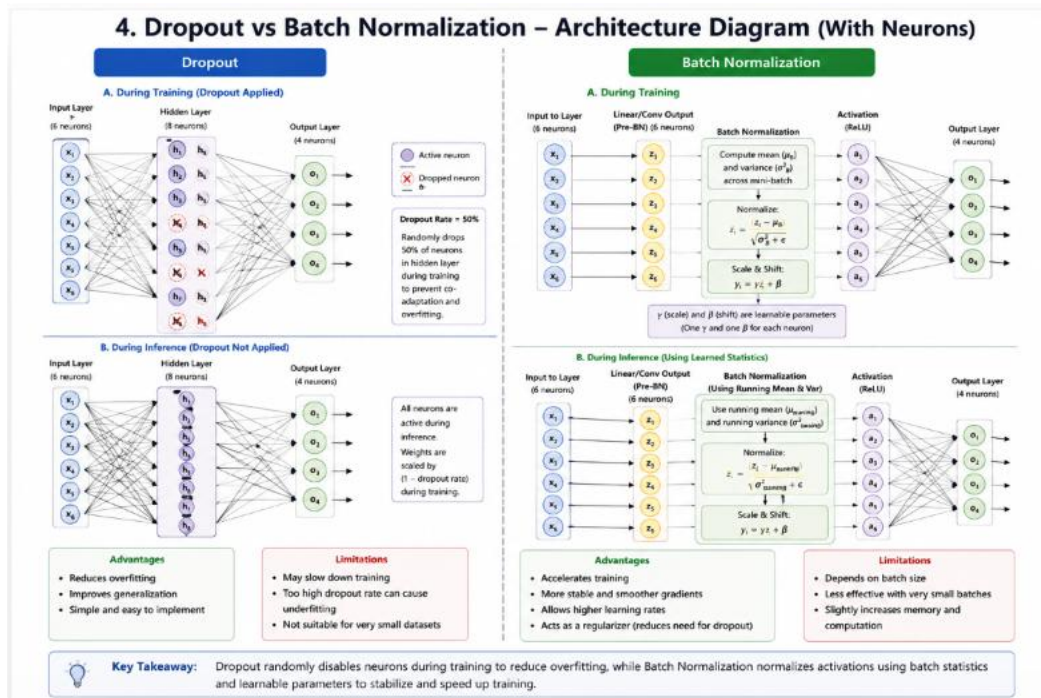
4. CNN Regularization Techniques

Explanation

Dropout randomly deactivates a percentage of neurons during training. This prevents the network from depending too much on specific neurons and reduces overfitting. During testing, all neurons are used. Dropout is simple and effective, but excessive dropout may reduce the learning capacity of the network.

Batch Normalization (BN) normalizes the activations of a mini-batch during training, making learning faster and more stable. Data Augmentation artificially creates new training examples using transformations such as rotation, flipping, cropping, scaling, and brightness changes. Data augmentation improves generalization by exposing the CNN to different variations of the same image.

ARCHITECTURE DIAGRAM



COMPARISON TABLE

Comparison Aspect	Dropout	Batch Normalization
Working	Randomly disables neurons	Normalizes layer activations
Main Purpose	Reduces overfitting	Stabilizes and speeds up training
Training	Random neurons are dropped	Uses mini-batch statistics
Testing	All neurons are active	Uses learned statistics
Advantage	Improves generalization	Faster and more stable learning
Limitation	May slow training	Performance can depend on batch size

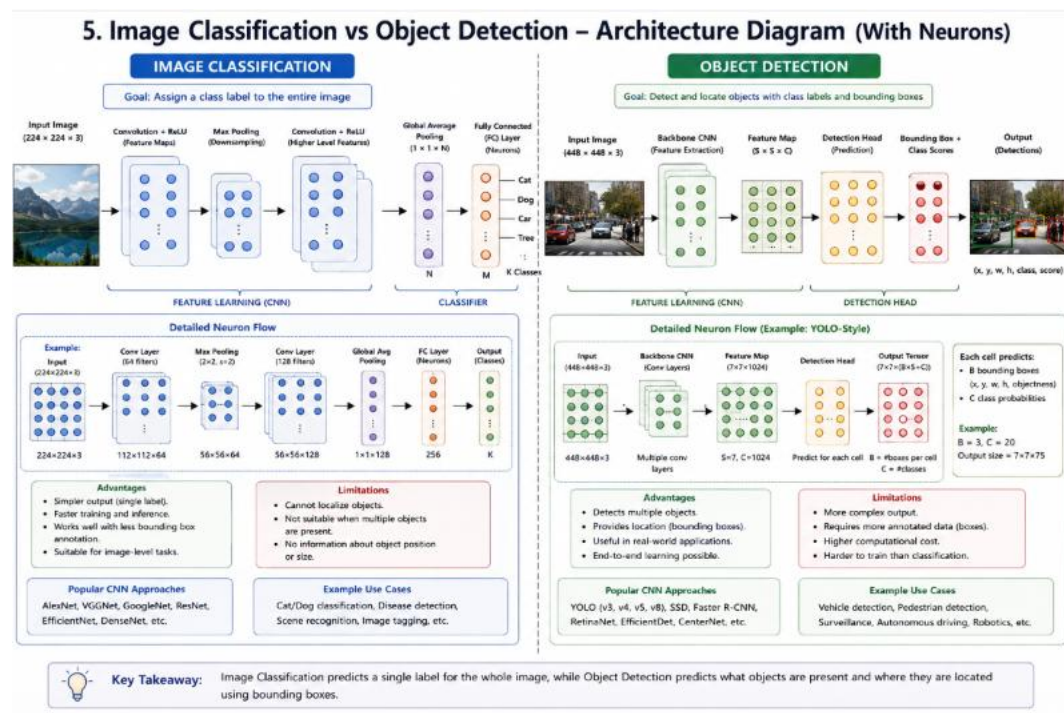
5. CNN Applications Analysis

Explanation

CNNs are widely used in image classification, object detection, face recognition, medical image analysis, and autonomous vehicles. In image classification, a CNN assigns a complete image to a class. Object detection goes further by identifying objects and locating them using bounding boxes. Face recognition uses CNNs to extract facial features and compare identities.

In medical imaging, CNNs can analyze X-rays, CT scans, MRI images, and other medical images to identify abnormalities. Autonomous vehicles use CNN-based systems to detect pedestrians, vehicles, traffic signs, lanes, and road conditions. Major challenges include large data requirements, lighting and environmental variations, occlusion, computational cost, and the need for high reliability.

ARCHITECTURE DIAGRAM



COMPARISON TABLE

Comparison Aspect	Image Classification	Object Detection
Purpose	Assigns a class to an entire image	Identifies and locates objects
Output	Class label	Class label + bounding box
CNN Approach	Classification CNNs such as ResNet	Detection CNNs such as YOLO
Complexity	Relatively simpler	More complex
Main Challenge	Similar-looking classes	Occlusion, scale and multiple objects
Application	Cat/dog classification	Vehicle and pedestrian detection